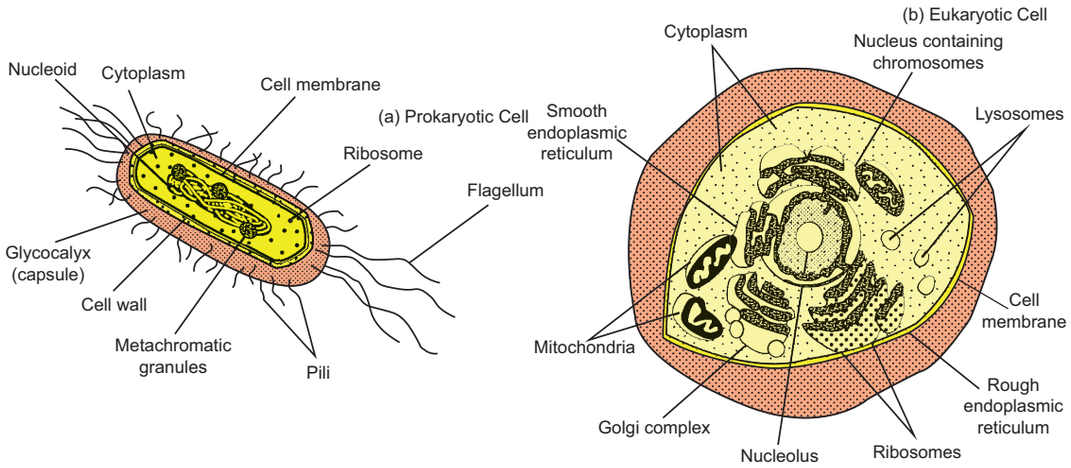


Figure 7.15: Structure of prokaryotic and eukaryotic cell

Table 7.6: Differences between prokaryotic and eukaryotic cell

S. No	Characteristic	Prokaryotic	Eukaryotic
1	Size of cell	Typically 0.2-2.0nm in diameter	Typically 10-100 nm diameters
2	Nucleus	No nuclear membrane or nucleoli	True nucleus, consisting of nuclear membrane and nucleoli
3	Membrane enclosed organelles	Absent	Present. Example: lysosomes, Golgi complex, endoplasmic reticulum, mitochondria and chloroplasts
4	Flagella	Consist of two protein building blocks	Complex, consist of multiple micro tubules
5	Glycocalyx	Present as a capsule or slime layer	Present in some cells that lack cell wall
6	Cell wall	Usually present and is chemically complex (typical bacterial cell wall includes peptidoglycan)	When present, chemically simple
7	Plasma membrane	No carbohydrates and generally lacks sterols	Sterols and carbohydrates that serve as receptors are present
8	Cytoplasm	No cytoskeleton or cytoplasmic streaming	Has cytoskeleton and shows cytoplasmic streaming
9	Ribosomes	70S	80S (70S in organelles)

(Continued)

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